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# What a Cheap Verifier Can and Cannot See: Telemetry Monitoring, Deterministic Checks, and Repair for Tool-Using LLM Agents

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## Abstract

1 Verifying an agent after it answers is too late, and verifying every step with a  
2 second model costs more than the agent. We ask what is verifiable from *observable*  
3 *step telemetry alone* — semantic embeddings of step output, token uncertainty,  
4 action metadata — by monitors that cost microseconds and are fitted on healthy  
5 runs only, and where that cheap signal provably runs out. On 2,823 committed  
6 episodes across three frameworks, four agent models and a commercial API, a  
7 one-class echo-state-network CUSUM ensemble detects  $0.71 \pm 0.07$  of failures at  
8 a 5% false-alarm budget. Two findings bound it. First, a *horizon law*: the temporal  
9 monitor’s advantage over a memoryless baseline grows from +0.09 detection  
10 points at  $\leq 3$  post-onset steps to +0.40 at  $\geq 9$ , and it predicts its own failure region  
11 on an external corpus we did not build: on AFTraj-2K, where 94% of failures end  
12 within eight steps of onset, pooled detection is 0.048 exactly as the law forecasts,  
13 while at  $\geq 9$  steps it is 0.509. Second, an *observability boundary*: a wrong-but-  
14 well-formed value perturbs no behavioural channel, so no reference-free monitor  
15 can see it. We therefore pair the monitor with deterministic checks that need no null  
16 and no threshold — recomputing a run’s stated total from the tool results it actually  
17 received — which catch 60% of failures at **0 of 63** false positives against the  
18 monitor’s 54% at 17%, transfer across model families unchanged (110/110 at 0/10),  
19 and close into repair that lifts task success from 52% to 73%. The contribution is a  
20 measured decomposition of which cheap observable verifies which failure, and an  
21 explicit statement of what neither verifies.

## 22 1 Introduction

23 An agent episode is a sequence of steps, each emitting observable telemetry: what the agent said, how  
24 confident its tokens were, which tool it called, what came back, how long it took. The verification  
25 question is whether that stream is enough to decide, *mid-run*, that something has gone wrong — early  
26 enough to halt, roll back, or repair.

27 Two answers dominate. Check the final output, which is cheap but arrives after the budget is spent; or  
28 judge every step with a second LLM, which is accurate and costs a model call per step. This paper  
29 measures a third position: a one-class monitor over step telemetry at  $\sim 200 \mu\text{s}$  per step, fitted on  
30 healthy runs only, with no failure labels and no second model. One optional variant — the logistic  
31 fusion of §5 — is supervised and needs on the order of twenty labelled failures; every other monitor  
32 and every check reported here is label-free. We are interested less in headline accuracy than in the  
33 *boundary* — what such a verifier structurally cannot see, and what to pair with it there.

## 34 Contributions.

- 35 1. **A horizon law with external validation.** The temporal monitor’s advantage over a memo-  
 36 ryless baseline is governed by post-onset runway, not by detector identity. The law predicts  
 37 where the monitor will fail, and holds on a corpus built by another group (§6). Only one of  
 38 our two external corpora can test it: the other labels trajectories rather than steps, so it has  
 39 no onset and no horizon.
- 40 2. **A measured observability boundary.** Content corruption that changes data without chang-  
 41 ing behaviour is invisible to behavioural telemetry; we show richer embeddings do not fix it,  
 42 and that a deterministic grounding channel does (§8).
- 43 3. **Deterministic verification that needs no null.** Checks that recompute a run’s answer  
 44 from the tool results it received beat the monitor on precision by construction — 0% false  
 45 positives against 17% — and transfer across model families with nothing retuned (§9).
- 46 4. **Verification closed into repair,** with the negative results kept: rolling a flagged run back  
 47 and re-running it lifts task success from 52% to 73%, but only one of five failure classes is  
 48 actually repairable (§9).
- 49 5. **A measured LLM judge,** not a stipulated one. The judge everyone escalates to scores  
 50  $p_{\text{detect}} = 0.548$  where analyses assume 0.90; substituting the measured value drops our own  
 51 escalation claim from 83% to 44% recovery, and we report the lower number (§7).

## 52 2 Telemetry and the Monitoring Problem

53 An episode is a sequence  $\mathcal{E} = \{(q_t, a_t, o_t, r_t)\}_{t=1}^T$  of reasoning  $q_t$ , action  $a_t$ , observation  $o_t$  and  
 54 runtime metadata  $r_t$ . At each step the adapter emits a causal feature vector

$$\mathbf{x}_t = [\mathbf{e}_t; \mathbf{u}_t; \mathbf{m}_t; \mathbf{g}_t] \in \mathbb{R}^d, \quad (1)$$

55 using only information available at or before  $t$ :  $\mathbf{e}_t$  a 32-dim deterministic character-trigram hash  
 56 of the step text (no model, no network),  $\mathbf{u}_t$  four token-surprisal aggregates where the provider  
 57 exposes logprobs,  $\mathbf{m}_t$  action type, latency, output length, retry and error metadata, and  $\mathbf{g}_t$  nine  
 58 content-grounding features (§8). Monitors are fitted on healthy episodes only and standardised with  
 59 healthy statistics,  $\tilde{\mathbf{x}}_t = \mathbf{D}^{-1}(\mathbf{x}_t - \boldsymbol{\mu})$ .

60 For a failed episode with onset  $\tau$ , a detector alarms at  $\hat{\tau} = \inf\{t : s_t > \gamma\}$ . We report detection at a  
 61 *realized* false-alarm rate rather than AUROC alone: a monitor that ranks well can still alarm after the  
 62 run ends or interrupt too many healthy runs, and both matter for a verifier that is allowed to halt.

## 63 3 Streams and the Serving Decision

64 No single detector observes every failure mechanism, so the system is a set of independently calibrated  
 65 streams.

66 **Temporal stream.** An ensemble of  $K=8$  leaky echo-state reservoirs [3] predicts the next standard-  
 67 ised step; member  $k$  has state  $\mathbf{h}_t^{(k)}$  and surprise  $e_t^{(k)} = \|\tilde{\mathbf{x}}_t - \hat{\mathbf{x}}_t^{(k)}\|_2$ . The ensemble exposes mean  
 68 surprise and disagreement, which are robustly  $z$ -scored and accumulated by a one-sided CUSUM [6],

$$C_t = \max\{0, C_{t-1} + z_t - \kappa\}, \quad (2)$$

69 so that weak but persistent deviations integrate rather than being smoothed away. Everything in (2)  
 70 depends only on  $\mathbf{x}_{0:t}$  and fit-time quantities: strictly causal, no lookahead.

71 **Abrupt-change stream.** Accumulation is useless when one or two steps remain after onset. A mem-  
 72 oryless  $\Delta$ -Mahalanobis stream scores healthy step differences  $\Delta\mathbf{x}_t$  against a shrinkage covariance [4],  
 73 and covers exactly the regime the temporal stream cannot (§5).

74 **The wrapper, not the reservoir, carries the margin.** The temporal stream is presented as an ESN  
 75 because that is what we ship, but the credit belongs elsewhere and the measurement says so. Giving  
 76 a GRU the same per-channel max-fusion wrapper lifts it to detection **0.76** at AUROC 0.873, past

the ESN’s 0.71 / 0.872; a linear vector-autoregression leads the ESN by roughly 1.9 steps of lead while trailing badly on detection. The transferable contribution is the wrapper — score each channel separately and take the max, rather than scoring a pooled vector. The reservoir stays the default for two practical properties rather than for accuracy: a lower healthy false-alarm rate (0.069 against 0.113) and a fit roughly 100× faster, which is what makes the per-deployment recalibration of §4 affordable at all.

**Calibration and the serving decision.** Each stream’s threshold is a quantile of per-episode maxima over *held-out* healthy episodes,

$$\gamma_j = Q_{1-\alpha_j} \left( \left\{ \max_t s_{j,t}^{(i)} : \mathcal{E}_i \in \mathcal{H}_{\text{val}} \right\} \right), \quad (3)$$

and the served alarm is a per-stream OR rather than a fused score,

$$A_t = \bigvee_j \mathbb{I}[s_{j,t} > \gamma_j] \vee B_t^{\text{contract}} \vee B_T^{\text{complete}}. \quad (4)$$

Equation (4) is a design result, not a formality. Under a single shared threshold only episode-max ordering matters, so the stream with the heavier healthy tail sets the bar for both: measured, one behavioural healthy episode destroyed **16 of 17** grounding-only detections of context corruption. No monotone rescaling fixes an ordering problem; independent thresholds do: no stream’s evidence can be numerically cancelled by another’s, which held at every seed we ran. The cost is paid in false alarms — an OR over  $J$  streams admits each stream’s budget — so this is a non-degradation property bought with alarm rate, not a free guarantee.

## 4 Real-Ecosystem Validation

Everything claimed about deployment rests on 2,823 committed episodes over 25 corpora — three frameworks, three local models across two families, and the gemini-2.5-flash API — each a live agent run against real tools, with telemetry measured at the adapter rather than constructed. Looping and tool cascade are caught almost everywhere (0.48–1.00) and goal drift wherever it is present (0.66–0.86 on four of five corpora); the weak spots are the content-corruption classes, `wrong_document` (0.00–0.20) and `malformed_json` (0.05–0.50), which is what motivates §8 rather than a rationalisation of it.

Two structural findings shape the rest. First, **monitors do not transfer across deployments**: calibrated on qwen2.5:7b and deployed on llama3.1:8b over a matched task, tool and injector plan, the primary monitor sits at chance (AUROC 0.527, healthy FA 0.75), while the *same* target recalibrated on itself reaches 0.885. That is a property of the calibration, not of a hard corpus, and it means a verifier of this kind is per-deployment by construction. Second, the memoryless baseline wins outright on short-episode datasets — which the next section explains.

## 5 The Horizon Law

Over 1,002 injected episodes the ESN – Mahalanobis detection gap is +0.09 at post-onset horizon  $\leq 3$  steps, +0.14 at 4–8 and +0.40 at  $\geq 9$  (correlation +0.25). The reservoir needs post-onset steps to integrate evidence: where it has them it wins decisively, and where it does not its margin collapses and the memoryless distance becomes competitive. This is a statement about *regime*, and it is what makes the monitor’s failures predictable rather than mysterious.

**The fusion buys robustness, not dominance.** A cross-fit logistic fusion of the two parents reaches grand-mean AUROC 0.826 against 0.802 (ESN) and 0.807 ( $\Delta$ -Mahalanobis), because each parent collapses on some dataset and the fusion never does. Measured against whichever parent is better *on that dataset*, however, it is at or below it on 7 of 8 datasets in AUROC (mean −0.014) and on all 8 in detection rate (mean −0.140).

The robustness that buys is real but conditional, and §6 contains its counterexample: on ATBench the memoryless parent is at chance (0.457) and drags every 50/50 blend down with it (0.438–0.463) against the ESN alone at 0.779, while on AFTraj the same fusion is marginally ahead (+0.015).

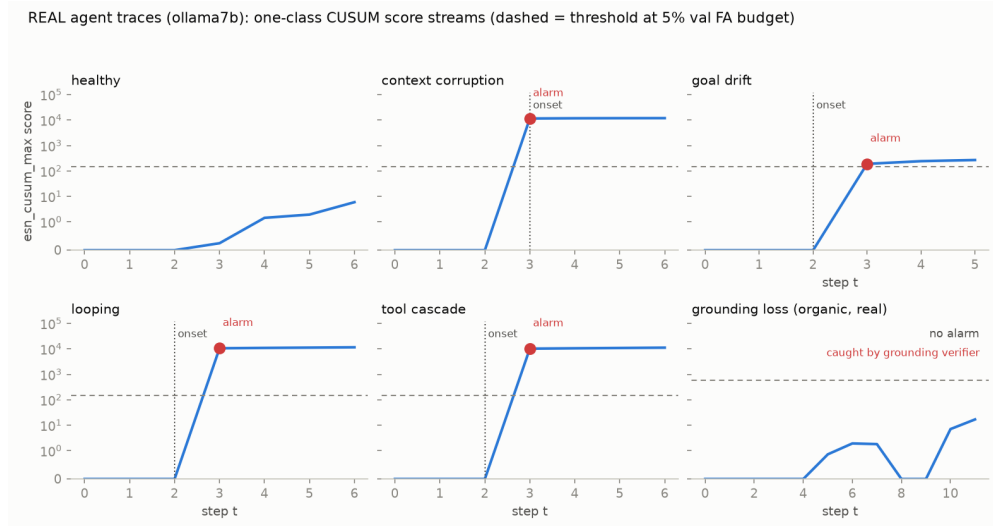


Figure 1: **What the verifier sees.** One-class CUSUM streams per failure class on real agent traces (qwen2.5:7b against real tools); dashed = threshold at the 5% validation budget, dotted = verified onset. Healthy stays two orders of magnitude below the line and every injected class alarms one step after onset. The sixth panel is different in kind: fabrication cannot be injected into a live run, so it is a genuine organic episode — the CUSUM stays flat, because a fabricated figure perturbs no behavioural channel, and the class is caught instead by the deterministic verifier of §9. That is the observability boundary shown rather than asserted. The  $y$  axis is symlog: the CUSUM accumulates multiplicatively once a failure takes hold.

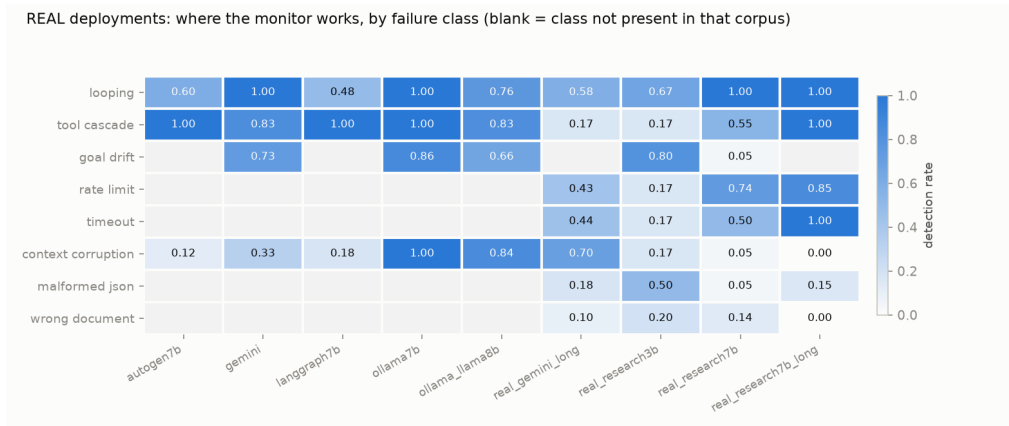


Figure 2: **Where the monitor works, by failure class**, across nine real corpora; blank where a corpus does not contain that class, because absent and undetected are different claims. Looping and tool cascade are caught almost everywhere and goal drift wherever present; the weak spots are exactly the content classes that motivate §8.

121 Fusion protects against *a* parent being weaker; it does not protect against one being uninformative,  
 122 because a 50/50 blend has no way to notice. The hybrid is a reasonable default when the regime is  
 123 unknown *and* both parents are known to carry signal — not a general safety net, and not a dominance  
 124 result.

## 6 External Validation on Corpora We Did Not Build

Every number above comes from corpora we collected, so none of them says whether the benchmark is hard. We therefore ran the monitors, unchanged and fitted on each corpus’s own healthy runs, against two external benchmarks.

**AFTraj-2K** [11] contributes 1,882 multi-agent trajectories after the same  $T \geq 4$  filter (1,111 healthy, 771 failed) with a judge-annotated earliest-decisive-error step. *Ranking transfers zero-shot*: episode AUROC **0.745** on another group’s agents and frameworks, with the uncertainty channel absent entirely, against 0.802 pooled on our own. *The horizon law predicted the operating point, and held*: detection is **0.509** on failures with  $\geq 9$  steps of runway against 0.170 for the memoryless baseline, but only 53 of 771 AFTraj failures have that much room and 61% end within three steps of onset, so pooled detection is **0.048**. That pooled figure is the mechanism’s prediction, not an excuse offered after seeing it: §5 fixed the relationship on our corpora, and AFTraj supplies a failure distribution concentrated where the relationship says detection is impossible. A law that forecasts its own failure region and then survives a corpus it never saw is stronger evidence than a higher pooled number would have been.

**ATBench** [8] contributes 604 trajectories (279 safe, 325 unsafe) labelled under a different taxonomy of risk source, failure mode and harm. Here the monitor *alarms*: AUROC **0.779** at detection **0.311**, because ATBench’s unsafe behaviour is a pattern of action — an over-privileged call, an unvalidated tool result — rather than a single wrong value. Two independent confirmations fall out. The fusion collapses to chance (0.457 for the memoryless parent, dragging every 50/50 blend to 0.438–0.463 against the ESN’s 0.779 alone), which is §5’s pooled-not-dominant finding shown on foreign data. And per failure mode the split is sharp — over-privileged actions 0.508 and unvalidated tool outputs 0.473 against inaccurate information **0.038** — reproducing, on labels we did not write, exactly the content blind spot §8 exists for.

For scale: an LLM auditor on AFTraj reports 66.44 Exact-F1 using a model call at every step [11]. We do not match that and do not claim to; detection at a false-alarm budget is not Exact-F1. The comparison we can make is cost: 173  $\mu$ s and no model call, against one 7B forward pass.

## 7 The Judge Is a Verifier Too, and It Should Be Measured

The default answer to “who verifies the agent” is another LLM. Escalation analyses — ours included — usually *stipulate* that judge’s operating point. We measured it instead. A gemini-2.5-flash judge on a labelled subset of 161 distinct prompts scores  $p_{\text{detect}} = \mathbf{0.548}$  (95% CI 0.44–0.65) and  $p_{\text{false}} = \mathbf{0.052}$ , against the 0.90/0.02 the analysis assumed; both stipulated values lie outside their measured intervals. Substituting the measured rates and re-running all five master seeds with everything else fixed, the *call saving survives* (6% of judge-every-step calls) but detection recovery falls from 83% to **44%**, supported at **0 of 5 seeds**. We report the escalation layer’s detection claim as 44%.

The layer is still worth having, for a reason that only appears once both are measured on the same episodes: judge and monitor fail on *different* classes (Figure 3). The judge is perfect on goal drift (21/21) and nearly blind on context corruption (0.18), where the monitor is the reverse (1.00). That is the empirical case for escalation as a complementary verifier rather than a more expensive approximation of the cheap one — and a caution against citing a stipulated judge as a baseline.

## 8 The Observability Boundary and the Grounding Channel

Behavioural monitors share one blind spot: corruption that changes *data* without changing *behaviour*. A wrong-but-well-formed number perturbs no channel in (1), so no reference-free detector can reject it. The *g* channel adds nine causal content features — query–result dissimilarity, result self-consistency, JSON-prefix validity, character anomaly, and a binary lexical-relevance flag costing 2  $\mu$ s — which makes query–evidence–reasoning *inconsistency* observable without claiming to observe truth. Pooled over 602 injected episodes, the content gate lifts content-class detection 0.23  $\rightarrow$  **0.40** and behavioural detection 0.73  $\rightarrow$  **0.80**, gaining 37 and 25 discordant detections against zero lost.

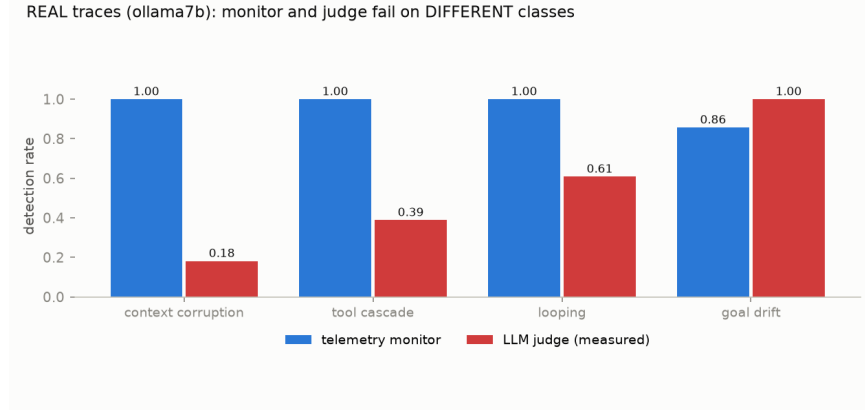


Figure 3: **Monitor and judge fail on different classes**, both measured on the same real corpus. The judge is perfect on goal drift and nearly blind on context corruption; the monitor is the reverse. Complementary verifiers, not substitutes.

173 The boundary is real and we measured its floor: on a plausible-corruption set, richer MiniLM  
 174 embeddings leave detection at **0.00** while behavioural AUROC falls 0.82 to 0.70, at 4.4 ms per text  
 175 and 90 MB. The failure is observability, not feature capacity — which is the argument for §9 rather  
 176 than for a bigger encoder.

## 177 9 Deterministic Verification, and Repair

178 The monitor needs a healthy null per deployment and spends part of a false-alarm budget. A  
 179 check needs neither. `total_consistency` recomputes a run’s stated total from the tool results  
 180 *that run actually received*; `required_coverage` confirms every call the task requires was made;  
 181 `tool_contract` asks whether a result matched any shape its tool can return, and so reports when  
 182 the result arrives rather than at the answer. None reads the hidden world the task was generated from.

Table 1: Checks against the behavioural monitor, same episodes and same objective labels. Recall is comparable; the difference is precision.

|                                   | checks                          | monitor     |
|-----------------------------------|---------------------------------|-------------|
| failures caught, $T=0.2$ (served) | 60% ( <b>96%</b> with coverage) | 54%         |
| false positives, $T=0.2$          | <b>0/63 = 0%</b>                | 11/63 = 17% |
| failures caught, $T=0.9$          | 65% ( <b>96%</b> with coverage) | 40%         |
| false positives, $T=0.9$          | <b>0/38 = 0%</b>                | 6/38 = 16%  |

183 The checks were written by inspecting failures in the serving arm, so that arm cannot also be their  
 184 test set. A further 120 episodes at disjoint task seeds were collected afterwards and scored with the  
 185 checks frozen: 54% caught by totals, **93%** with coverage, and **0 of 64** false positives. A llama3.1:8b  
 186 arm on the *same* task seeds, with nothing retuned, catches **110 of 110** at **0 of 10** — the checks  
 187 transfer where the monitor does not, because they carry no calibration. Across every labelled corpus  
 188 `tool_contract` trips on **0 of 1825** healthy episodes and flags 46% of context corruption, **215 of 218**  
 189 within one step of onset.

190 **Repair.** Each flagged episode is rolled back to its last fact-gathering step and re-run, paired on  
 191 the identical prefix and task; the rollback is real, and success is graded by an oracle the repair  
 192 prompt never sees. Naming which check failed — without supplying any value — recovers **45%** of  
 193 genuinely-wrong runs against a 16% resampling control ( $p = 0.0005$ ), which nets task success from  
 194 **52% to 73%** with zero correct runs broken. Two negative results sit inside that: routing the step to a  
 195 calculator the agent already holds does *not* beat retry luck ( $p = 0.17$ ), and supplying the recomputed  
 196 answer buys nothing — 26 of 55 value-bearing hints contain the correct total outright, while the  
 197 value-free rung recovers at least as much. The recovery does not come from handing over the answer.

Coverage is partial and the shape of the gap is measured: over five injection classes  $\times$  five seeds, every behavioural alarm is followed by a repair attempt (21 of 21), but `goal_drift` is the only class a retry fixes (4 of 5). Where the tool layer itself is broken a retry fetches the same broken result, and the value of the intervention is ending the episode fast — a loop trap exits at exactly 10 steps against 30 without the circuit breaker.

## 10 Limitations

Stated as measured. (1) **No cross-deployment transfer**: the healthy null must be collected under the exact serving distribution; transferred across a model family the monitor sits at chance (0.527) while the same target recalibrated reaches 0.885. (2) **Slow goal drift** evades every per-step surprise monitor tested; the limit is the *rate* of change, not the class — abrupt goal changes are caught at 0.66–0.86. (3) **Content detection is bounded by telemetry completeness**, not by logprob access: ablating the token-surprisal channel moves the ESN by AUROC +0.000. (4) **Plausible-value corruption** is undetectable from telemetry by construction. (5) **Fabrication claims are limited by base rate**: 9 of 175 organic episodes are labelled hallucinated and only 2 are fabricated *inputs*, below the pre-registered minimum of 10, so no detection claim is made in either direction; under provocation the grounding verifier catches 0.55 with no healthy false positives — specific, but half sensitive. (6) **Scope**: mock-tool and research-loop tasks, two local model families and one commercial API.

## 11 Related Work

**Concurrent online failure prediction.** Detecting failure during an episode is no longer an unoccupied position. AgentForesight reframes attribution as online auditing and releases AFTraj-2K [11]; weakly supervised early alerting learns turn-level risk from trajectory labels alone [1]. Both use an LLM auditor. We therefore do not claim mid-episode detection from observable telemetry as novel; what is ours is the cost class and the boundary analysis.

**Internals versus observables.** Hidden-state probes report predicting failure earlier than any observable-behaviour monitor [7]. We do not dispute it: with the weights, activations are richer. The premise here is the deployment where you do not hold them — a hosted API, a third-party agent — and there the choice is between telemetry and nothing.

**Runtime enforcement** intervenes during execution but requires a specification of what unsafe means: rule sets [9] or labelled unsafe states [10]. Ours is label-free and specification-free, which is why it transfers badly and needs no rule authoring. **Post-hoc attribution** asks which step caused a failure after the fact and finds it hard — 14.2% accuracy on the responsible step [12]; that problem is outcome-conditioned, ours is not. **Hallucination detection** needs model access or repeated sampling [5, 2]; tool-using agents admit the cheaper deterministic route of §9.

## 12 Conclusion

A cheap verifier over step telemetry covers more of the agent-failure space than its cost suggests, and its failures are predictable rather than arbitrary: the horizon law says where it will not fire, and held on a corpus built by another group. Where it structurally cannot see — a wrong-but-well-formed value — the answer is not a bigger encoder but a different mechanism, and a deterministic check that recomputes the answer from the evidence the run received is both more precise and more portable than the monitor it supplements. We think the useful framing for agent verification is coverage: which cheap observable verifies which failure, measured, with the gaps stated.

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## A Appendix

### A.1 Telemetry versions

| Version | Dims | Content                                                                                                                                       |
|---------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| v1      | 43   | $e_t$ 32-dim hash embedding; $u_t$ 4 token-surprisal aggregates; $m_t$ action one-hot, log latency, output length, error                      |
| v2      | 43   | Tool results appended into step text so corrupted results reach the semantic channel                                                          |
| v3      | 51   | 8 derived dims: cosine drift, task-anchor similarity, tool success, retry, per-tool latency, context ratio, reasoning depth, self-consistency |
| v4      | 60   | 9 content-grounding dims: query–result dissimilarity, result self-consistency, JSON-prefix validity, character anomaly, lexical flag          |

Table 2: Causal telemetry evolution. v4 costs 608  $\mu$ s/step at the adapter; the lexical flag alone costs 2  $\mu$ s/result.

### A.2 Per-dataset hybrid results

### A.3 External corpora

### A.4 Reproducibility

Code, all 2,823 committed traces, every result table and the figures are publicly released under a permissive licence; the URL is withheld for review. Each headline number in this paper is tied to



| dataset              | ESN   | $\Delta$ -Maha | better parent | logistic | $\Delta$ |
|----------------------|-------|----------------|---------------|----------|----------|
| sim                  | 0.890 | 0.786          | 0.890         | 0.889    | -0.000   |
| autogen7b            | 0.833 | 0.774          | 0.833         | 0.777    | -0.056   |
| langgraph7b          | 0.828 | 0.885          | 0.885         | 0.884    | -0.002   |
| ollama7b             | 0.994 | 0.895          | 0.994         | 0.982    | -0.013   |
| real_research3b      | 0.556 | 0.665          | 0.665         | 0.643    | -0.022   |
| real_research7b      | 0.777 | 0.848          | 0.848         | 0.847    | -0.001   |
| real_research7b_long | 0.790 | 0.849          | 0.849         | 0.857    | +0.008   |
| gemini               | 0.749 | 0.756          | 0.756         | 0.731    | -0.025   |
| grand mean           | 0.802 | 0.807          | 0.840         | 0.826    | -0.014   |

Table 3: Episode AUROC. The fusion beats either parent’s grand mean while sitting below the per-dataset better parent on 7 of 8.

| corpus    | monitor               | AUROC | detection | FA    |
|-----------|-----------------------|-------|-----------|-------|
| AFTraj-2K | esn_cusum_max         | 0.745 | 0.048     | 0.023 |
| AFTraj-2K | hybrid_weighted50     | 0.760 | 0.047     | 0.018 |
| ATBench   | esn_cusum_max         | 0.779 | 0.311     | 0.071 |
| ATBench   | $\Delta$ -Mahalanobis | 0.457 | 0.268     | 0.161 |
| ATBench   | hybrid_weighted50     | 0.463 | 0.277     | 0.179 |

Table 4: Monitors run unchanged on two external corpora, fitted on each corpus’s own healthy runs. Neither carries token logprobs, so both run on  $e+m+x$  only.

278 a named artifact by a claim ledger that recomputes it from the committed file on every run, and a  
279 SHA-256 manifest covers every source file, trace and result so a reader can verify the number in the  
280 text came from the file in the repository. An end-to-end behavioural snapshot re-runs the study at a  
281 fixed seed and compares every value against a stored baseline.